

Breaking the Link

Final Solution and Reflection on the Development Process

Data Visualization Project — Deliverable 4

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Live dashboard: <https://breaking-the-link-economic-decoupling-dataviz.streamlit.app/>

Repository: github.com/kristine-p/economic-decoupling-dataviz

Demo video: <https://www.youtube.com/watch?v=h6miKog9yLs>

All links also available in the footnotes

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1. Introduction

This report constitutes the fourth and final deliverable of the Data Visualization course project Breaking the Link, an interactive, map-based dashboard investigating the decoupling of economic growth from greenhouse gas (GHG) emissions. Its purpose is twofold: to present the final visualization solution, and to document and critically discuss the development process that produced it. In particular, the differences between the final solution and the original proposal (Deliverable 1), the reasons for those differences, and the difficulties encountered during development.

The central research question has remained unchanged since the proposal: **can a nation expand its economy while simultaneously shrinking its environmental footprint?** The means by which this question is addressed, however, evolved substantially. The proposal outlined a traffic-light world map, a toggleable air-quality layer, and a scenario slider; all three concepts are present in the final product, but each was significantly rethought and redesigned during development. The final map employs the complete eight-class Tapio classification, still conveyed by means of a traffic-light design, but with more in-depth insight; the air-quality layer displays 723 sub-national regions rather than one symbol per country; and the scenario view is built on a data-driven linkage between present decoupling behavior and projected climate futures rather than on a split-screen comparison. Section 4 documents each of these changes and their reasoning, addressing one of the core requirements of this deliverable.

The report is organized as follows. Section 2 presents the final solution, including its architecture, the three analytical layers, and its design system. Section 3 reconstructs the development flow in seven phases, since several design decisions can only be properly understood in the order in which the underlying criticalities emerged. Section 4 systematically compares the final solution with the proposal and provides a justification for each detected deviation. Section 5 discusses the difficulties encountered during development, grouped into data engineering, geospatial enrichment, visualization interaction, and framework limitations, together with the measures adopted to address them. Section 6 summarizes the conclusions that arise from the project.

2. The Final Solution

2.1 Overview

Breaking the Link is a fullscreen, web-based dashboard implemented in Python with Streamlit and Plotly, publicly deployed on Streamlit Community Cloud. The interface is organized around a single edge-to-edge world map, over which all controls float as translucent panels: a navigation element at the top, a control panel on the left, a country detail panel on the right, and a timeline dock at the bottom. This layout, inspired by the live view of [Electricity Maps](#), constitutes a deliberate departure from Streamlit's default document-style layout; the map is treated as the primary object of attention, with all remaining elements functioning as instrumentation around it.

The dashboard is supported by a reproducible preprocessing pipeline (panel.py) that integrates four sources: OECD greenhouse gas emissions, World Bank GDP per capita

(PPP), OECD PM2.5 exposure, and OECD climate projections. The integration produces three processed tables: a merged country panel across 90 countries (depending on the view) covering 1990–2023 (2,420 country-year observations); a sub-national PM2.5 table of 723 regions across 52 countries (16,629 observations, enriched with geocoded centroid coordinates); and a climate projection table covering 52 countries, four SSP scenarios, and the years 2030–2060 (6,448 observations). The core derived variable is the Tapio decoupling elasticity ($E = \frac{\Delta\%GHG}{\Delta\%GDP}$) (Tapio, 2005; Wang et al., 2022) computed both annually and as a five-year rolling mean, and classified according to Tapio's typology of seven decoupling states, complemented by an explicit 'undefined' class for years with near-zero GDP growth or missing data.

2.2 View 1: the Global Decoupling Map

The entry point of the dashboard is a choropleth map in which each country is colored according to its Tapio classification for the selected year. The three traffic-light categories sketched in the proposal got enriched and became a seven-class Tapio states classification, grouped in the legend under the headings 'GDP growing' and 'GDP contracting'. Within each class, the exact shade encodes the country's actual elasticity value on a per-class gradient, so that two countries sharing a classification remain visually distinguishable according to their position within it. A timeline slider allows the period 1990–2023 to be explored, and a smoothing control switches between the annual index and its five-year rolling average, which absorbs yearly distortions such as the one introduced by the COVID-19 shock of 2020. The right-hand panel displays a yearly summary with the share of tracked countries currently decoupling, together with the fully annotated legend. The default year presented on first load is not the most recent year available but the most recent year retaining at least 80% of peak country coverage (2022): World Bank GDP reporting lags by one to two years for many economies, and the unadjusted latest year would present a substantially incomplete map.

2.3 View 2: the air-quality ('Human Breath') layer

A toggle in the left panel overlays population-weighted PM2.5 concentrations on the Decoupling Map (View 1). This layer underwent the largest change in granularity relative to the proposal: instead of a single proportional symbol per country, the final version plots 723 sub-national regions, each geocoded to its administrative centroid and rendered as a two-pass 'haze', a wide translucent halo beneath a solid, readable core, on a WHO-inspired clean-to-hazardous color scale deliberately disjoint from the Tapio palette. While the layer is active, the underlying choropleth is dimmed so that the two encodings do not compete, and the timeline is automatically restricted to 2000–2020, the interval for which PM2.5 data exist. The regional granularity is what renders the layer analytically meaningful: it exposes intra-country contrasts, e.g. between heavily industrialized and rural areas of the same country, that a single national average necessarily conceals, and it is precisely these contrasts that connect the macroeconomic narrative to population-level air quality.

2.4 View 3: Future Scenarios

The forward-looking view maps projected changes in three extreme-weather indicators: hot days (maximum temperature above 35 °C), tropical nights (minimum above 20 °C), and icing days (maximum below 0 °C) under the four SSP scenarios, from 2030 to 2060. The map

encodes each country's projected percentage change in days per year relative to its own 1981–2010 historical baseline, on a diverging blue-to-red scale centered exactly on zero: warm hues where the indicator increases relative to the baseline, cool hues where it decreases. The baseline-relative difference, in means of percentage change, was preferred over the absolute projection because absolute day counts predominantly encode intrinsic geographical attributes, whereas the change against a country's own history encodes the quantity of interest, namely the pace of climatic disruption.

The defining design decision of this view is the scenario linkage. By default the map does not display a single global scenario. Instead, each country is drawn under the SSP pathway linked to its current Tapio classification: countries in absolute decoupling under SSP1-2.6, relatively decoupling economies under SSP2-4.5, coupled economies under SSP3-7.0, and negatively decoupling economies under SSP5-8.5. The map therefore functions as a conditional statement: the projected climate a country is approaching if its present behavior persists. The linkage can be overridden through a dropdown selecting any single SSP scenario, and selecting a country, either by clicking it on the map or through a synchronized dropdown, populates a trend chart in the right panel showing all four scenario trajectories for that country to 2060, with the linked pathway drawn as a solid line and the alternatives dotted, so that the gap between the best and worst pathway remains permanently visible.

2.5 Design system and interaction model

All visual constants reside in a single shared module (`style.py`): a light 'instrument panel' theme comprising a soft grey background, white translucent panels, and a signal-teal accent reserved exclusively for interactive controls, so that interface chrome never competes with the three data palettes (the Tapio green-to-terracotta scheme, the PM2.5 haze scale, and the diverging projection scale). Panel positions are governed by CSS custom properties rather than independently chosen pixel values, preventing the side panels and the timeline dock from drifting out of alignment. Interaction follows a single consistent grammar across all views: the bottom dock always controls time, the left panel always controls what is encoded, the right panel always provides detail on a selected country, and the map itself is always pannable, zoomable, and hoverable and, in the Future Scenarios view (View 3), clickable.

2.6 Technical implementation and reproducibility

The codebase is modular: `app.py` manages layout and application state; `view1.py`, `view2.py`, and `view3.py` each implement one analytical layer; `style.py` contains the design system; `.streamlit/config.toml` sets the application theme, including the primary green accent used for interactive controls such as the timeline slider; and `panel.py` together with `geocode.py` constitute the data preparation stage. Interface behavior is covered by headless tests written against Streamlit's AppTest harness, exercising tab switching, toggles, selectors, and inspection of the rendered HTML. The repository documents a complete replication path, consisting of raw data acquisition instructions, a single-command pipeline run, an optional geocoding step, and dashboard launch. Also, the geocoded coordinates are committed to the repository as a static file specifically so that replication does not depend on the current state of the OpenStreetMap database. The dashboard is publicly deployed on Streamlit Community Cloud.

3. Development Flow

Development proceeded in seven phases, reconstructed in *Table 1*. The order of these phases is relevant, not just for completeness: several of the most substantial design changes documented in Section 4 were direct consequences of problems identified in earlier phases, and the rationale for those changes is only fully intelligible against this chronology.

Phase	Focus	Summary
P0	Proposal (D1)	Definition of topic, audience, and the three-view architecture; specification of the traffic-light encoding, the 'Human Breath' overlay, and a split-view scenario slider; formulation of an OECD-centric data plan.
P1	Data acquisition and pipeline v1	Acquisition of OECD GHG and World Bank GDP data; wide-to-long transformation, exclusion of aggregate codes, correction of encoding artifacts; first computation of the Tapio elasticity and its eight-class classifier. The GDP source was changed from the proposed OECD output-approach table to World Bank GDP per capita PPP on grounds of country coverage.
P2	First prototype	First choropleth implemented in Streamlit's default layout with a year slider; the three-class traffic-light scheme was implemented and immediately found to conceal analytically relevant distinctions (Section 4.1).
P3	Data-integrity incident and pipeline v2–v3	The join of PM2.5 data silently inflated the panel elevenfold and rendered nearly all Tapio classifications undefined; the root cause was traced to the source file's eleven-rows-per-country-year structure; a dual filter (MEAN_POP with EXPOSURE_LEVEL = _Z) and mandatory key-uniqueness assertions were introduced. The panel was rebuilt on a broader GHG download, expanding coverage from approximately 42 to 90 countries.
P4	Interface redesign	Migration from the default document layout to the fullscreen, HUD-style interface: fixed translucent panels, pill navigation, a CSS design-token system, and the resolution of several conflicts with Streamlit's own page structure (Section 5.5).
P5	Regional PM2.5 and geocoding (pipeline v4)	Identification of 723 sub-national regions encoded within the PM2.5 source file; addition of a region-level pipeline branch; development of a resumable Nominatim geocoder with a manually curated override map of more than sixty entries, achieving 100% coordinate coverage; reconstruction of View 2 around regional symbols with haze rendering and choropleth dimming.
P6	Future scenarios (pipeline v5)	Integration of the climate projections (~1.5 GB) via chunked reading; addition of baseline-difference measures and re-specification of the map from absolute day counts to change relative to 1981–2010; implementation of the per-country Tapio-to-SSP linkage; replacement of Plotly's native selection machinery with a two-trace opacity pattern for country selection (Section 5.4).
P7	Consolidation	Test coverage with AppTest; introduction of the coverage-anchored default year; refinement of legends and tooltips; completion of the methodology dialog, README replication path, and public deployment; preparation of this report.

Table 1: The seven phases of development, from proposal to deployed dashboard.

4. Differences Between the Final Solution and the Proposal

The final solution differs from the proposal in a number of aspects. The structural commitments of the proposal were preserved in full: the research question, the target audience (policymakers, environmental consultants, and the informed public), the medium (an interactive Streamlit dashboard designed for desktop use), the three-layer architecture progressing from a global overview through human exposure to future scenarios, and the principle that optional layers must be toggleable to prevent visual overload. Within that structure, however, most design decisions were revisited during development. *Table 2* summarizes the differences; the following subsections provide the justification for each.

Aspect	Proposal (D1)	Final solution
Map encoding	Traffic-light scheme: three categories (dark green / yellow-orange / orange-red)	Complete Tapio typology: seven states plus 'undefined', with a per-class gradient encoding the exact elasticity value
Scenario view	Option A: split-view map comparing two scenarios side by side (primary sketch)	Option B, refined: per-country SSP linkage derived from the current Tapio class, complemented by a four-scenario trend chart for the selected country
Scenario metric	Projected values under SSP scenarios	Change in days per year relative to each country's 1981-2010 baseline, on a diverging scale centered on zero
Air-quality layer	One proportional symbol per country	723 geocoded sub-national regions; two-pass haze rendering; dimming of the underlying choropleth
Country detail	Hover-triggered data overlays	Click-to-select on the map (Future Scenarios view only) + Hover for additional info
GDP source	OECD GDP (output approach, PPP)	World Bank GDP per capita PPP (indicator NY.GDP.PCAP.PP.CD)
Coverage	OECD-centric (approximately 40 countries implied)	90 countries in the historical panel; 52 in the projections; 52 in the regional PM2.5 layer
Time range	'Up until 2024'	1990-2023, with the default year anchored to best data coverage (2022)
Normalization	Separate 'intensity ratios' (pollution per unit of GDP)	Subsumed by the Tapio elasticity itself; per-capita GHG retained as tooltip context
Layout	Unspecified (implicitly Streamlit by default)	Fullscreen map with floating translucent panels and a shared CSS design-token system

Table 2: Summary of differences between proposed and final designs.

4.1 From the traffic-light scheme to the complete Tapio typology

The proposal specified a three-color traffic-light scheme intended to maximize interpretability: green for decoupling, red for coupling. The first prototype implemented this scheme, and

testing revealed a deficiency that had not been anticipated: the reduction to three classes can be actively misleading. Tapio's framework contains three states in which GDP is contracting, and under the traffic-light scheme a country in recession whose emissions decline faster than its economy would receive the same green coloring as a country achieving genuine absolute decoupling under growth. The message 'green equals success' would therefore have been factually incorrect other than misleading for such countries. This became particularly evident for the year 2020, in which pandemic-induced recessions caused a large share of the map to turn green for reasons unrelated to sustainable growth. The final map consequently distinguishes all seven states, following the complete Tapio (2005) typology, with the legend explicitly partitioned into 'GDP growing' and 'GDP contracting' groups so that this essential context is never suppressed.

This decision raised the objection articulated in the proposal itself, namely that seven categories may overburden an audience without an economics background. The objection was addressed through three measures, in preference to reducing the information content: the legend annotates every class with a one-line plain-language description; an information dialog explains the complete framework, including the formula, on demand; and hue families preserve the traffic-light intuition: green tones for decoupling, warm orange and earthy red tones for coupling, so that the glance-level message survives while the detail-level message becomes more accurate. A second encoding channel, absent from the proposal, was additionally introduced: within each class, a gradient maps the country's actual elasticity value within the range, preserving ordinal information that a flat categorical fill would discard.

4.2 The scenario view: from split-screen comparison to data-driven linkage

The proposal presented two candidate designs for View 3: a split-view map comparing two scenarios for a fixed year, and a data-driven 'default' scenario selection linking each country's present Tapio classification to the most plausible future pathway. Prototyping began with the former, but the latter was adopted, for three reasons. First, the split view imposes a spatial comparison task, which both the perception literature and internal testing indicate is considerably more difficult than reading a single map whose condition is toggled; the equivalent comparison remains available in the final design by switching the scenario dropdown. Second, the split view offers no principled answer to the question of which scenario each half should display, whereas the linkage transforms the view from a neutral atlas of possible futures into a coherent argument connecting present behavior to projected consequences, which constitutes a considerably stronger narrative bridge from Views 1 and 2. Third, rendering two synchronized choropleths doubled the figure cost and multiplied the interaction defects described in Section 5.4.

The linkage required one methodological caveat, which the dashboard's information dialog states explicitly: the mapping from Tapio classes to SSP scenarios is a narrative device rather than a causal model, since the decoupling performance of a single country does not determine a global emissions pathway. The gain in rhetorical clarity was judged to outweigh this simplification, provided that it is disclosed and that the user retains the ability to dissolve the linkage and inspect any scenario directly, both duties explicitly performed by the dropdown and the four-line chart. A further change within the same view concerns the color encoding, which was moved from absolute projected day counts to the change relative to each country's own 1981-2010 baseline: during testing, the absolute map largely reproduced a map of latitude, with warm-climate countries appearing alarming and cold-climate countries

appearing safe irrespective of their individual rates of change. The baseline-relative diverging scale, centered on zero, displays the intended signal that is the pace of disruption.

4.3 Data-driven changes: sources, coverage, and granularity

Three changes were either necessitated or enabled by the data itself. First, the GDP source was changed from the proposed OECD output-approach table to World Bank GDP per capita PPP, whose country coverage is considerably broader. Second, the processed climate projection table ultimately covers 52 countries rather than the 38 OECD members initially extracted, narrowing the coverage asymmetry recorded as a known limitation in the technical report; 38 countries of the historical panel remain without projections and are rendered as no-data areas in the Future Scenarios view (View 3). Third, the OECD PM2.5 file was found to encode sub-national regions alongside country aggregates, distinguishable by whether the first portion of the region code coincides with the country ISO code; an earlier pipeline version had silently discarded these rows. Retaining them increased the layer's spatial resolution by an order of magnitude and is the reason the Air Quality layer (View 2) displays intra-country pollution structure rather than a single symbol per country. Finally, the proposal's separate 'intensity ratio' derivative was subsumed rather than abandoned: the Tapio elasticity is itself the ratio that renders small and large economies comparable, and per-capita emissions remain available as tooltip context, so a dedicated intensity encoding would have duplicated information already present.

4.4 Interaction model and temporal honesty

The proposal intended hover as the primary gesture for detailed exploration. The final interaction distinguishes hover for transient tooltips from click for persistent selection: clicking a country in the Future Scenarios view (View 3) selects it, synchronizes the dropdown, and populates the trend panel. Separately, the proposal's commitment to a timeline extending 'up until 2024' presupposed data that is not yet available; the final panel ends, accurately, at 2023, the PM2.5 layer at 2020 (with the timeline automatically restricted while the layer is active), and the default year is selected based on recency and data availability.

5. Difficulties Encountered

The proposal anticipated three categories of challenge: data granularity, visual overload, and data integration. What had not been anticipated is that the most consequential problems would be silent: not failures that halt execution, but failures that produce a plausible-looking yet incorrect dashboard. This section groups the difficulties into four areas and, for each, describes the symptom, the diagnosis, the resolution, and the lesson derived.

5.1 Silent data corruption in the PM2.5 join

The formative technical incident of the project occurred shortly after the PM2.5 data were joined into the panel: the map turned predominantly grey, with 88% of country-year observations classified as 'undefined'. No error was raised, the Tapio pipeline simply ceased to produce classifications. The cause lay three steps upstream of the symptom. The OECD PM2.5 file contains eleven rows per country-year: one mean-concentration measure and ten WHO threshold-band shares, and the unfiltered join duplicated every panel row elevenfold.

On duplicated rows, the percentage-change computation returns zero; a zero GDP change activates the near-zero-growth guard introduced to keep the elasticity ratio numerically stable and the guard, functioning as designed, rendered every observation undefined. Each component behaved correctly in isolation, but their composition did not. The immediate resolution was a dual filter (MEASURE = MEAN_POP and EXPOSURE_LEVEL = _Z), the durable resolution was procedural: the pipeline now asserts key uniqueness after every load and merge operation, so that this class of defect fails immediately at its source rather than manifesting downstream as an unexplained visual anomaly. A structurally analogous defect was present in the GHG file, in which CO2-only rows duplicate the GHG totals unless three filters (measure, unit, and pollutant) are applied simultaneously. The newly introduced assertions detected this case before it reached the visualization.

5.2 Processing at scale: the climate projection file

The raw OECD climate projection export is approximately 1.5 GB, exceeding the capacity of a single in-memory read on the available hardware. The file was therefore processed in chunks of 100,000 rows, with each chunk filtered to country-level observations, the four SSP scenarios, and the six retained measures (three absolute projections and their published differences against the 1981-2010 baseline) before concatenation. Two concerns emerged: First, isolated non-numeric values in the observation column could silently coerce the entire concatenated column to a string type; each chunk is therefore defensively treated as numeric, with invalid rows dropped and logged. Second, when percentage change was derived from the baseline differences, an initial 'baseline too small' threshold of 1.0 silently suppressed cold-climate countries whose hot-day baselines are legitimately small. Tightening the threshold to 0.01 restored these countries while continuing to guard against the threat of division by zero. The episode demonstrated that every threshold in a data pipeline constitutes a modeling decision with substantive consequences rather than a formality.

5.3 Geocoding 723 sub-national regions

The regional PM2.5 layer required coordinates that the OECD data do not provide, since the source contains only region codes and names. A geocoder was constructed around the Nominatim (OpenStreetMap) API, and immediately encountered the divergence between administrative-statistical and local geographic nomenclature: OECD labels carry suffixes such as '(TL2021)' that no gazetteer recognizes, and generic English designations fail to resolve: 'North West', for instance, returns no result, while the correct local name for the corresponding Bulgarian region is 'Severozapaden'. The resolution comprised systematic label cleaning, normalization of geocoder-incompatible country names (such as 'China (People's Republic of)' and 'Türkiye'), and a manually curated override map of more than sixty entries translating OECD labels into locally recognized equivalents, assembled by investigating each failure individually. Because Nominatim's usage policy limits requests to approximately one per second, a complete run requires roughly fifteen minutes and must tolerate interruption; the geocoder therefore persists its progress incrementally after every lookup and resumes from partial output. Final coverage is 100% of the 723 regions. Two consequences are recorded. As a limitation, Nominatim returns administrative rather than population-weighted centroids, so symbols may be displaced from the areas where exposed populations are concentrated in large or unevenly populated regions; this is disclosed in the

documentation. As a reproducibility decision, since OpenStreetMap results evolve over time, the coordinates are committed to the repository as a static file rather than regenerated at runtime, so reproducibility in this instance was achieved by freezing an external dependency rather than by re-executing it.

5.4 Selection interaction on choropleth maps

The most persistent interaction difficulty concerned the implementation of country selection by clicking on a choropleth. Plotly's native selection machinery (selected points with selected and unselected styles) appears designed for this purpose but proved unreliable in practice: selection styling compounded with the per-country opacity into double-dimming artifacts, selections intermittently required multiple clicks to register, and per-point opacity arrays introduced further rendering defects. After several reimplementations, the native machinery was abandoned in favor of a pattern relying exclusively on tools the choropleth renders reliably: a background trace containing all countries at low scalar opacity and a highlight trace containing the selected country at full opacity, with the click event serving only to reassign countries between traces. The same pattern was subsequently applied to Decoupling Map's dimming behavior when the PM2.5 overlay is active, providing independent confirmation of its robustness. One residual defect could not be fully resolved: deselection by clicking static regions, such as the ocean.

5.5 Adapting Streamlit to a fixed full screen layout

Streamlit is designed to render a scrolling document, whereas the design called for a fixed fullscreen canvas with floating overlays; reconciling the two required intervention at several levels. Three difficulties specific to overriding Streamlit's layout assumptions are worth recording: First, raw CSS injected through `st.markdown` was corrupted by the Markdown parser: indented rules were re-interpreted as code blocks and leading asterisks consumed as list markers. The fix was a single line: switching to `st.html`, which bypasses the Markdown pass entirely, but diagnosing the root cause took disproportionate effort. Second, Streamlit's header and wrapper elements continued to reserve layout space even when invisible, producing a persistent blank strip above the map. The resolution combined `display:none` (rather than `visibility:hidden`) on the framework chrome, collapse of the fixed top bar's block-flow wrapper, responsive rather than fixed figure sizing, and staggered synthetic resize events prompting Plotly to re-measure after the layout stabilizes. Third, CSS transforms or filters on any ancestor element silently re-anchor `position:fixed` descendants, displacing every floating panel, the stylesheet therefore defensively neutralizes these properties on the framework's wrappers, using broad selectors in preference to enumerating internal identifiers that change between Streamlit versions. The general conclusion is that where a framework's assumptions must be overridden, the override should be applied at the most general and version-stable level available, and every workaround should be documented with the failure it prevents, several of these measures being unintelligible without such documentation.

5.6 Statistical distortions and their treatment

A final family of difficulties derived not from tooling but from the relationship between the data and the phenomena they measure. The year 2020 produces extreme Tapio values for most countries, since simultaneous collapses of GDP and emissions render the elasticity ratio unstable, this is visible in the exploratory analysis as a spike of recessive classifications. The

five-year rolling average was consequently promoted from a secondary option to a core control; the annual view deliberately remains available, anomalies included, since suppressing 2020 would constitute a distortion of its own. The near-zero-GDP guard (elasticity undefined where $|\Delta\%GDP| < 0.01$) protects the ratio from numerical instability at the cost of an honestly labeled 'undefined' class on the map. Finally, the reporting lag of World Bank GDP data demonstrated that 'the latest year' is not a neutral default: the freshly loaded map appeared substantially incomplete until the default was anchored to the most recent year retaining at least 80% of peak coverage. Each of these decisions is individually small, but collectively they define the character of the dashboard, which indicates where the data are weak rather than concealing it.

6. Conclusion

Measured against the proposal, the final solution preserves every structural commitment, from the research question to the three-layer architecture, while revising nearly every concrete design decision beneath them, in each case for reasons this report has sought to make explicit: the traffic-light scheme was replaced because it could mislead; the split view because it worsened rather than supported comparison; the narrative annotations because they could not be executed credibly at the project's final scale; and the country-level symbols because the data proved to contain a representation an order of magnitude richer.

The data support a cautiously optimistic answer to the research question: a nation can indeed grow while reducing its environmental footprint, as documented by the map's progressive shift toward absolute decoupling from 2010 onward, but the pattern is neither universal nor irreversible. Air quality improvements lag behind reductions in emission accounts: populations still breathe polluted air even where emissions are falling on paper. Meanwhile, the climate projections show what current trajectories lead to if left unchanged.

The project ultimately reinforced a lesson central to data visualization: presenting data honestly requires as many decisions about what not to show, and how to qualify what is shown, as it does technical skill in rendering it.

7. References

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Data sources: OECD Data Explorer, World Bank Open Data.
Report prepared as Deliverable 4 for the Data Visualization course.